

- 1. In recent years, alliances and partnerships in aerospace and defense have been an historical foundation of the Department of Defense. What are some of the issues present in the C-17 case that highlight the difficulties of execution in multiple A&D business partnerships (see Chapters 10-11 in Thompson et al.)?**

The C-17 case illustrates several challenges associated with executing multiple aerospace and defense (A&D) business partnerships:

1. **Management Deficiencies:** The C-17 program's struggles with poor management underscore the importance of assembling a competent leadership team, as discussed in Chapter 10. Effective management is crucial for overseeing contracts and subcontractors, preventing schedule delays, and accurately gauging project scope. Chapter 10 emphasizes the significance of talent acquisition and building intellectual capital in various industries. I believe acquiring the right talent and providing adequate training could have potentially minimized the reliance on subcontractors for the C-17 program, and perhaps even rendered their hiring unnecessary. Even if they had to hire subcontractors, with the right expertise in place, the program might have been able to select the most suitable subcontractors, thereby reducing the likelihood of delays and integration problems.

2. **Technical Complexity:** The complexity of developing the C-17 aircraft demonstrates the necessity of clear technical specifications and effective coordination among partners, aligning with the insights from Chapter 11. I personally don't think subcontracting such a big responsibility was a good decision. Subcontracting out significant portions of software development without specifying a standardized programming language led to technical challenges. In my programming experience, using more than three programming languages for a

project can be very challenging in the long term. While it might seem like a quick fix for short-term issues, it often complicates matters down the line. Code is typically written with long-term usage in mind, as it needs to be understood and maintained by others in the future. Having multiple programming languages can increase the likelihood of the software or project being discarded in the future, as it may be easier to start anew rather than navigate through complex codebases. Chapter 11 covers the importance of recognizing the role of information systems and operating systems in enabling company personnel to carry out their strategic roles proficiently. Clearly, C-17 failed to recognize the importance of the information system.

3. Risk management : In Chapter 10, it is discussed that managers must accurately identify strategy-critical resources and capabilities. In the case of the C-17 project, concurrency, a high-risk development approach, can be considered one of these strategy-critical resources. Although concurrency was intended to accelerate innovation, its adoption introduced complexity and increased the likelihood of errors and delays. I think the use of firm-fixed price contracts in the C-17 project exemplifies the need for thorough risk assessment, as it can lead to cost overruns and unfulfilled contractual requirements if not managed properly. This demonstrates the significance of aligning risk management practices with strategy execution principles outlined in Chapter 10.

2. The C-17 development follows a typical new product development life cycle. The technology proved difficult to achieve at first but over time the aircraft has met most, if not all, of the Air Force's specification requirements. What does that say about innovation in

general, and, more specifically, the development of complex products? Is this another example of Finkelstein's "Holy Grail" phenomenon?

I believe the development of the C-17 aircraft reflects a common pattern seen in new product development cycles. Initially, the technology was challenging to master, leading to setbacks and difficulties. However, over time, through perseverance and innovation, the aircraft has successfully met most, if not all, of the Air Force's specifications. This experience highlights a broader lesson about innovation. It shows that innovation often involves overcoming initial obstacles and challenges. Despite facing difficulties, continuous efforts and improvements can lead to successful outcomes in the end.

Specifically, in the development of complex products like the C-17, this process is particularly evident. The intricate nature of such projects often requires significant time, resources, and expertise to navigate. However, as seen with the C-17, diligent efforts and advancements in technology can eventually lead to meeting or exceeding the desired specifications. I see strong resemblance between the journey of the C-17 development and life itself. It's about persistently pushing forward, even when pieces don't fit together until it does by making mistakes and learning from it.

I personally did not find this C-17 example as the "Holy Grail" phenomenon, I think it's not quite fitting. The Holy Grail concept typically refers to the pursuit of an unattainable goal that holds great attraction but ultimately remains elusive. In the case of the C-17, while there were challenges and setbacks, the project ultimately succeeded in meeting its objectives. Therefore, I believe it's more of a testament to the power of perseverance and innovation rather than a pursuit of an unattainable ideal.

3. Why would DAC want this contract? It seems that a firm-fixed priced (FFP) contract is not the right type for this type of development. How do organizational contracts like this affect transaction costs of a firm in general? Using FFP contracts in A&D development and production seems like an application of the “Wrong Scoreboard” situation Finkelstein describes. Agree or disagree? Why?

The reasons DAC would want this contract is because:

1. Competitive advantage: If we look from business perspectives, winning a FFP contract enhances DAC's reputation and competitiveness in the aerospace and defense industry, leading to future contract opportunities. It is a big win to even gain trust with the project. This is a way to stay ahead of their competitors.
2. Predictable revenue stream: As we saw from the passage FFP contracts provided DAC with a clear and guaranteed payment structure, reducing financial uncertainty. I believe this implies maybe they will not cover the cost more but they could try to finish it within the fixed cost which is a guaranteed project for the company.

Organizational contracts like FFP contracts can affect transaction costs for a firm in several ways. Firstly, I think the most crucial one is they can increase the administrative burden and monitoring costs associated with ensuring compliance with the contract terms. Secondly, in cases where the project scope changes or unforeseen challenges arise, renegotiating the contract terms can be time-consuming and costly. Thirdly, FFP contracts may incentivize contractors to limit innovation to fit in the initial fixed cost constraints, compromising the quality of the final product. Overall, I think FFP contracts can drive up a firm's transaction costs.

"Wrong Scoreboard" typically refers to a situation where individuals or organizations are measuring success or progress using the wrong criteria or metrics. Yes, I think using FFP contracts in A&D development and production seems like an application of the "Wrong Scoreboard" situation.

In the case of the C-17 program, the adoption of FFP contracts was driven by specific objectives such as cost overruns and perceiving low risk due to the integration of commercial off-the-shelf (COTS) technology. I think this is where the idea of the wrong scoreboard comes in because this is a wrong way to measure the success. This narrow focus could potentially lead to detrimental outcomes, as cost-saving measures may compromise essential aspects of the project's success. For instance, subcontracting decisions made solely to reduce costs may not prioritize the expertise and capabilities required for optimal project execution, resulting in unforeseen expenses and delays. Therefore, while FFP contracts address certain aspects of project management, they risk neglecting other vital metrics necessary for achieving favorable outcomes in A&D development and production.

On the other hand, I think we can also argue it is not a wrong score board because issues arose due to poor management, lack of domain knowledge, and technical complexities, rather than a misalignment of metrics or goals. Therefore, while the use of FFP contracts may have contributed to certain challenges in the C-17 development, it does not necessarily represent a "Wrong Scoreboard". Instead, I believe it highlights the importance of effective management, expertise, and risk assessment in A&D partnerships.

4. Finally, it appears that many of the suppliers in this case including DAC, HC and GE did not have the subject matter expertise (i.e., think knowledge, skills and abilities (KSA)) to

successfully develop, test and manufacture the C-17 aircraft. What does this say about the value of human resources in successfully implementing an organization's strategy?

I believe this proves the importance of human resources in successfully implementing an organization's strategy. Firstly, the significance of hiring individuals with the right expertise cannot be overstated. For example, leading companies like Google, Deloitte, and Netflix prioritize talent acquisition, investing resources in rigorous selection processes to ensure they onboard individuals with the necessary knowledge and skills. I believe in the case of the C-17 program, if the suppliers possessed the requisite subject matter expertise, it's plausible that many of the challenges encountered could have been mitigated. This emphasizes the pivotal role of having employees with the appropriate KSA, as it directly impacts the organization's ability to navigate complex projects successfully.

Secondly, even if the company doesn't hire people who have subject matter expertise a company could always train the employees as long as they have the ability to do so. The importance of ongoing training and development should not be overlooked. I believe training would not be a good example in this case of the C-17 project as it wasn't company employees but sub contractors that lacked the subject matter expertise. But other organizations can leverage training initiatives to enhance the KSA of their workforce.

Although, knowledge and skills are very important in work life. According to my experience, these companies such as Google, Deloitte, and Netflix also focus too much on it more than ability. I believe they prioritize knowledge and skills more than ability, which may not always be good for companies because I believe the main important key aspect in a competitive

candidate would be ability. As long as the candidate has the ability to do some tasks knowledge and skills are gainable.

Lastly, I believe the absence of subject matter expertise among suppliers in the C-17 program serves as a compelling illustration of the critical value of KSA within an organization, highlighting the importance of recruiting, retaining, and developing employees with the requisite knowledge, skills, and abilities to drive strategic success.